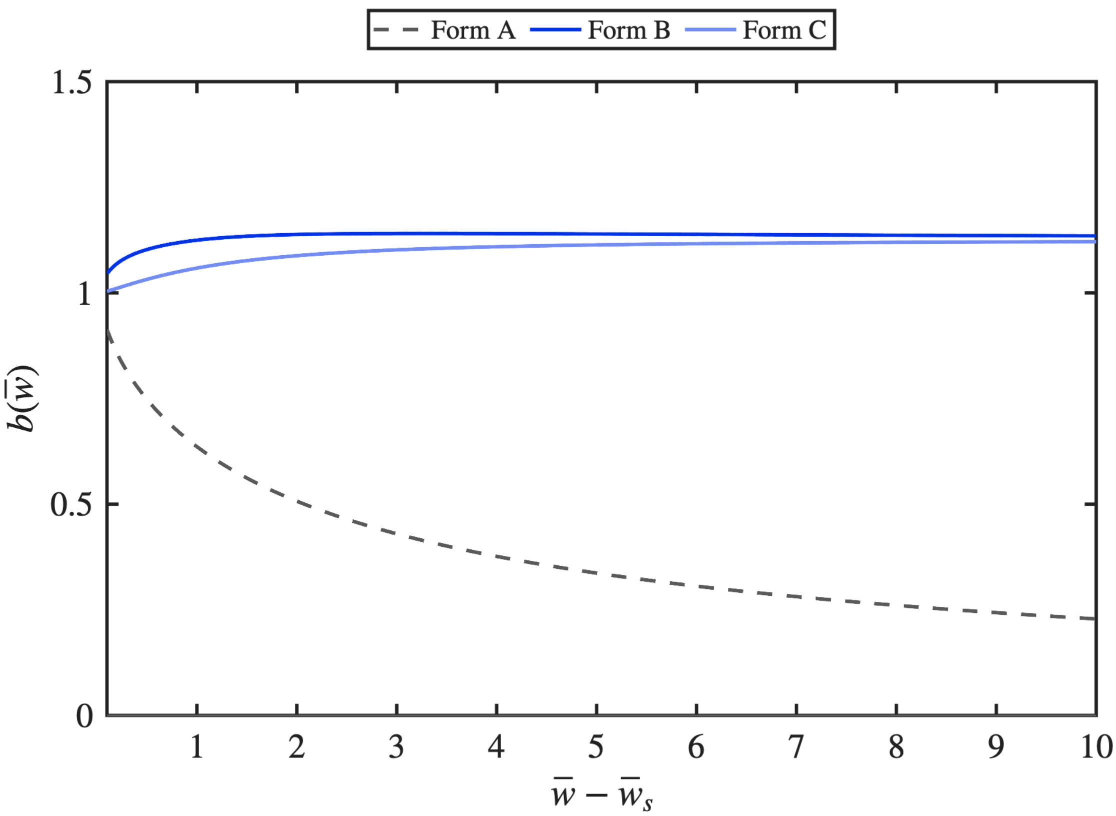
$$\begin{aligned}
&\text{Form A} \quad \bar{a} = 1 - e^{-b\left(\bar{w}-\bar{w}_s\right)} && p = 0 \\
&\text{Form B} \quad \bar{a} = 1 - \frac{b}{\left(\bar{w}-\bar{w}_s\right) + b} && p = 1 \\
&\text{Form C} \quad \bar{a} = 1 - \frac{b}{1 + \bar{w} - \bar{w}_s} && p = 1
\end{aligned}$$

$$\bar{w}_s \text{ known}, \qquad \bar{a}'_{\text{true}} := \frac{\mathrm{d}\bar{a}_{\text{true}}}{\mathrm{d}\bar{w}}$$

$$b_{\mathrm{A}}(\bar{w}) = -\frac{\ln\left(1-\bar{a}_{\text{true}}\right)}{\bar{w}-\bar{w}_s}, \qquad b_{\mathrm{B}}(\bar{w}) = \left(\bar{w}-\bar{w}_s\right)\frac{1-\bar{a}_{\text{true}}}{\bar{a}_{\text{true}}}, \qquad b_{\mathrm{C}}(\bar{w}) = \left(1+\bar{w}-\bar{w}_s\right)\left(1-\bar{a}_{\text{true}}\right)$$

$$\bar{w}-\bar{w}_s \in [\,0.1,\ 10\,]$$

$$\begin{aligned}
\bar{w}-\bar{w}_s = 0: &\quad \text{A, B}: \text{ any } b\,; \qquad \text{C}: \ b = 1 \\
\bar{w}-\bar{w}_s = \infty: &\quad \text{A, B, C}: \text{ any } b \\
\text{elsewhere}: &\quad \partial\bar{a}/\partial b \neq 0 \ \Rightarrow \ b(\bar{w}) \text{ unique}
\end{aligned}$$

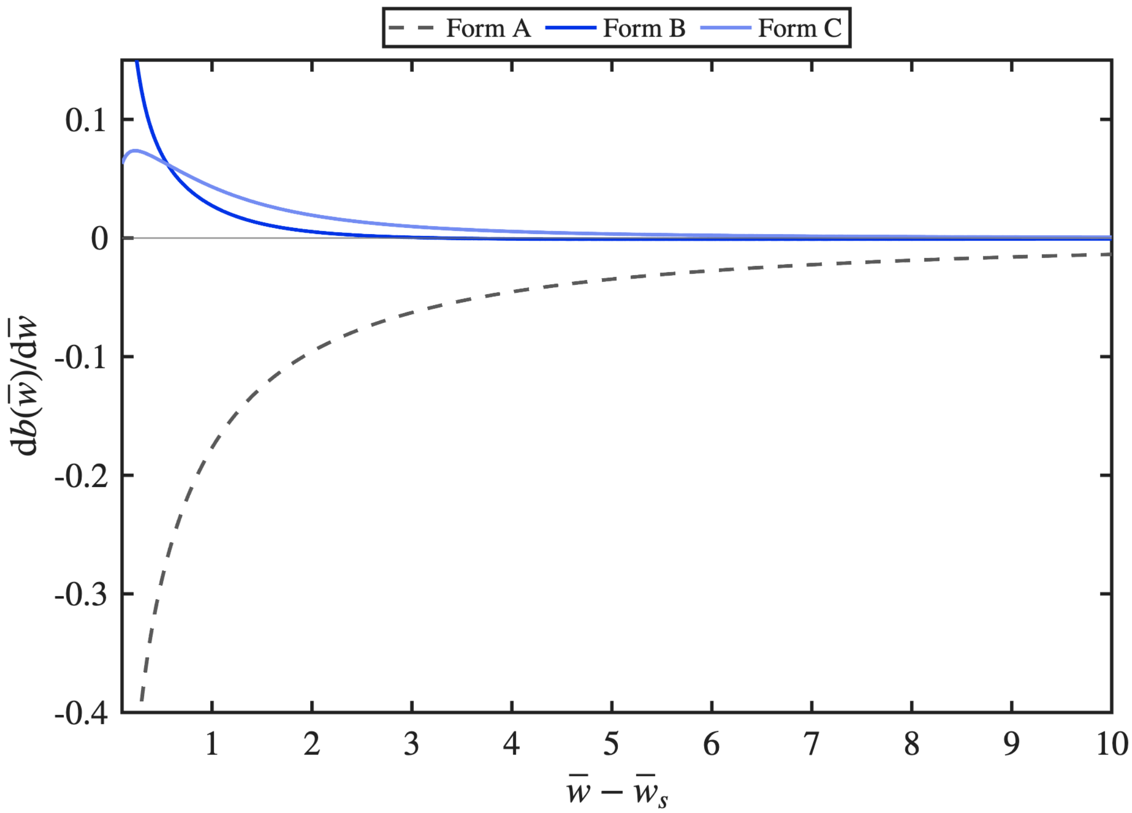


$$\frac{\mathrm{d}b_{\mathrm{A}}(\bar{w})}{\mathrm{d}\bar{w}} = \frac{1}{\left(\bar{w}-\bar{w}_s\right)^2}\left[\frac{\left(\bar{w}-\bar{w}_s\right)\bar{a}'_{\mathrm{true}}}{1-\bar{a}_{\mathrm{true}}} + \ln\left(1-\bar{a}_{\mathrm{true}}\right)\right]$$

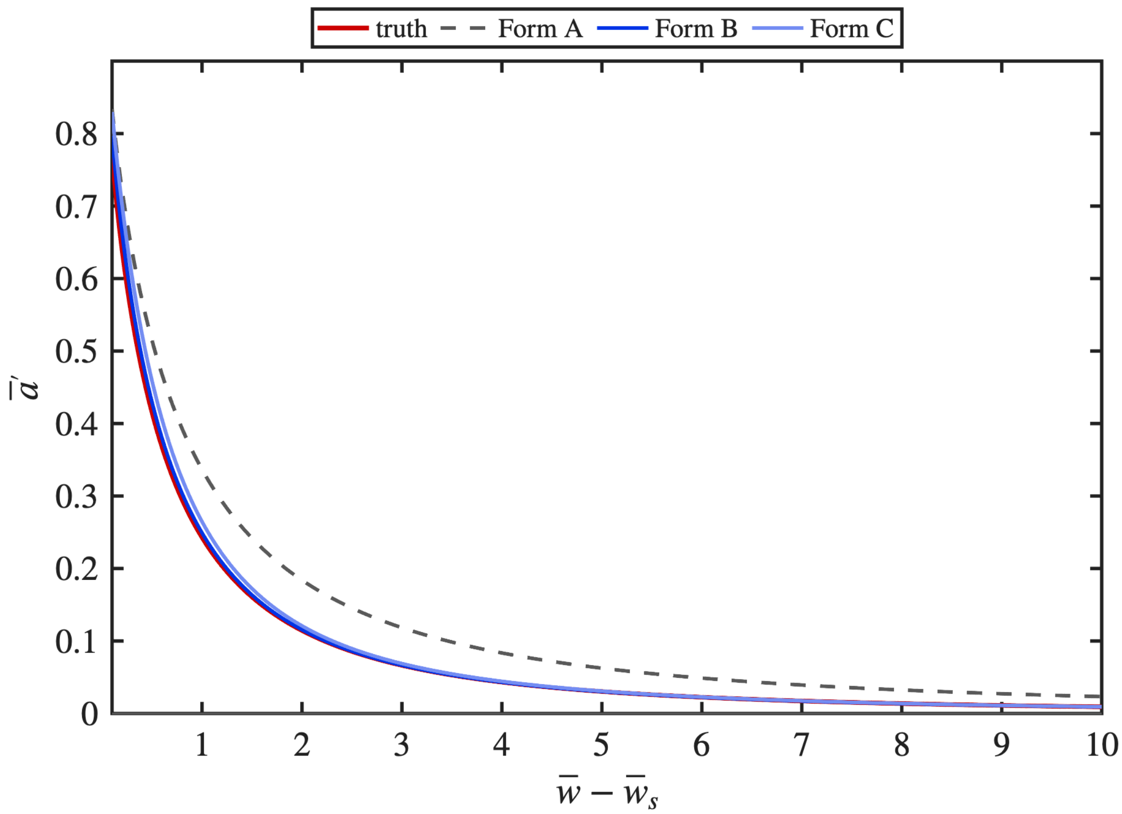
$$\frac{\mathrm{d}b_{\mathrm{B}}(\bar{w})}{\mathrm{d}\bar{w}} = \frac{1-\bar{a}_{\mathrm{true}}}{\bar{a}_{\mathrm{true}}} - \left(\bar{w}-\bar{w}_s\right)\frac{\bar{a}'_{\mathrm{true}}}{\bar{a}^2_{\mathrm{true}}}$$

$$\frac{\mathrm{d}b_{\mathrm{C}}(\bar{w})}{\mathrm{d}\bar{w}} = \left(1-\bar{a}_{\mathrm{true}}\right) - \left(1+\bar{w}-\bar{w}_s\right)\bar{a}'_{\mathrm{true}}$$

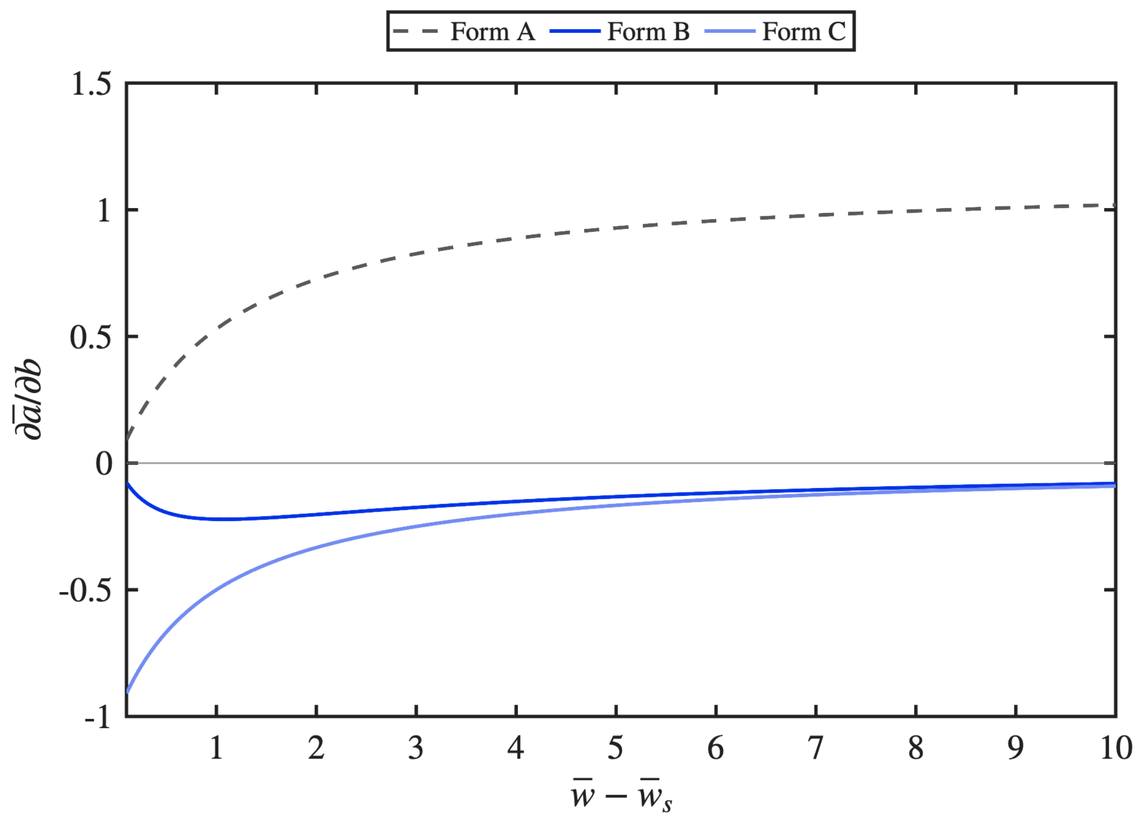
$$\left.\frac{\mathrm{d}b(\bar{w})}{\mathrm{d}\bar{w}}\right|_{\bar{w}-\bar{w}_s=0.1} = -0.669,\ +0.316,\ +0.063 \qquad \left.\frac{\mathrm{d}b(\bar{w})}{\mathrm{d}\bar{w}}\right|_{\bar{w}-\bar{w}_s=10} = -0.014,\ -0.001,\ +0.001$$



$$\bar{a}'_{\mathrm{A}} = -\frac{\left(1-\bar{a}_{\mathrm{true}}\right)\ln\left(1-\bar{a}_{\mathrm{true}}\right)}{\bar{w}-\bar{w}_s} \qquad \bar{a}'_{\mathrm{B}} = \frac{\bar{a}_{\mathrm{true}}\left(1-\bar{a}_{\mathrm{true}}\right)}{\bar{w}-\bar{w}_s} \qquad \bar{a}'_{\mathrm{C}} = \frac{1-\bar{a}_{\mathrm{true}}}{1+\bar{w}-\bar{w}_s}$$

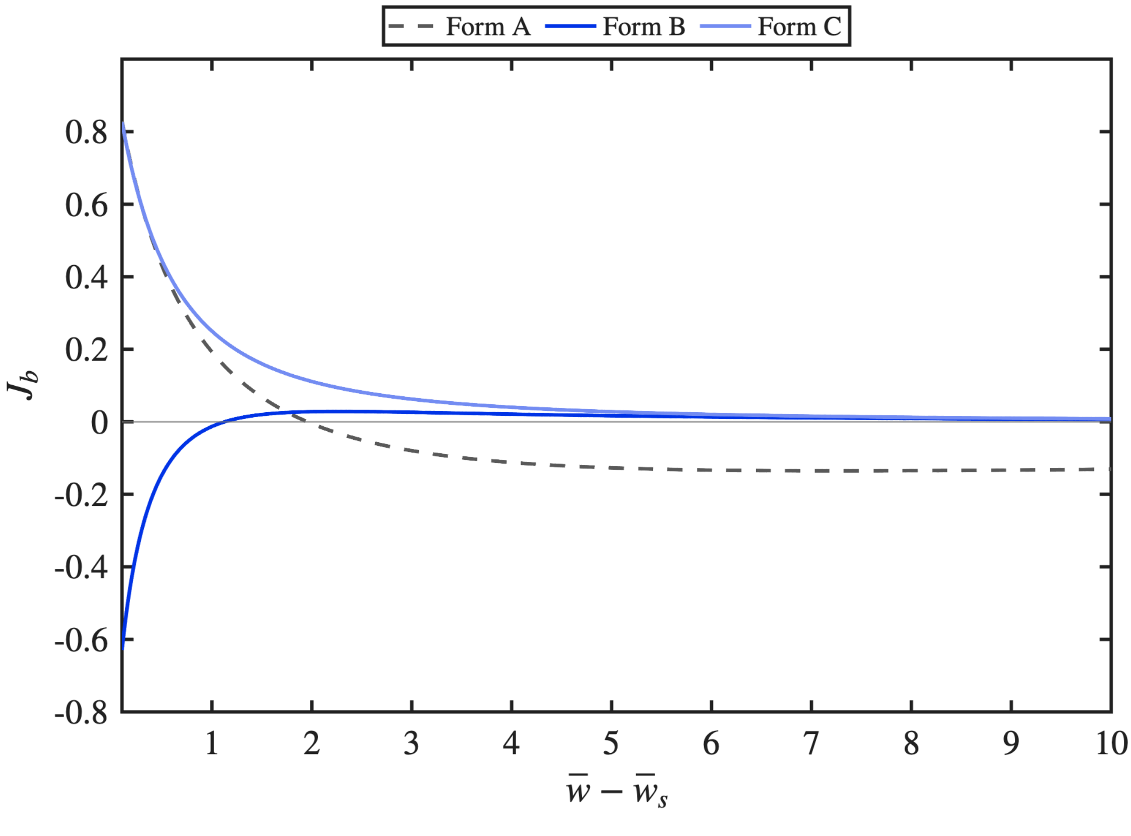


$$\frac{\partial \bar{a}_{\mathrm{A}}}{\partial b} = +\big(\bar{w} - \bar{w}_s\big)\big(1 - \bar{a}_{\mathrm{true}}\big) \qquad \frac{\partial \bar{a}_{\mathrm{B}}}{\partial b} = -\frac{\bar{a}_{\mathrm{true}}^2}{\bar{w} - \bar{w}_s} \qquad \frac{\partial \bar{a}_{\mathrm{C}}}{\partial b} = -\frac{1}{1 + \bar{w} - \bar{w}_s}$$



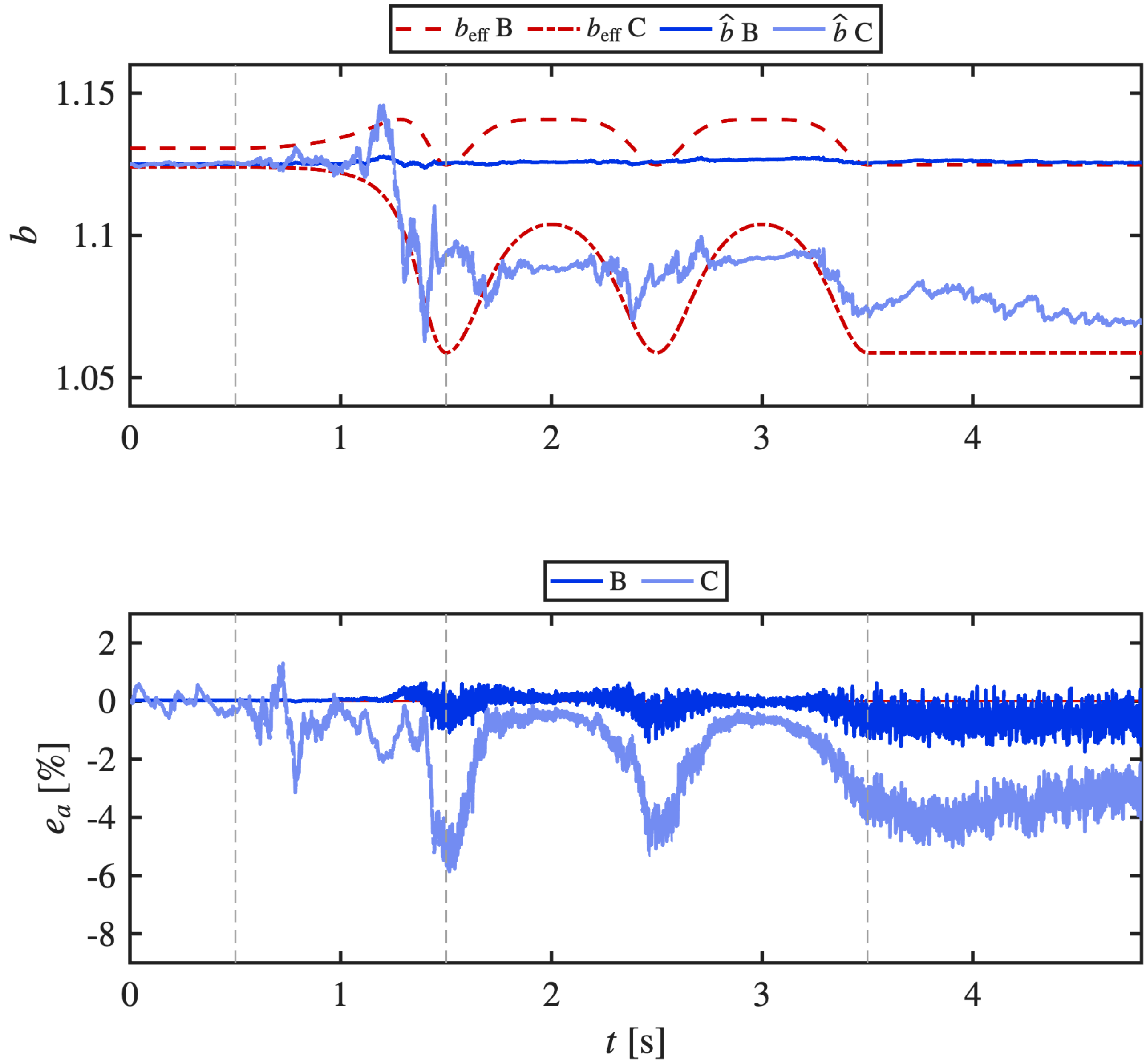
$$J_{b,\mathrm{A}} = \big(1-\bar{a}_{\mathrm{true}}\big)\Big[1+\ln\big(1-\bar{a}_{\mathrm{true}}\big)\Big] \qquad J_{b,\mathrm{B}} = \frac{\bar{a}_{\mathrm{true}}^2\big(2\bar{a}_{\mathrm{true}}-1\big)}{\big(\bar{w}-\bar{w}_s\big)^2} \qquad J_{b,\mathrm{C}} = \frac{1}{\big(1+\bar{w}-\bar{w}_s\big)^2}$$

$$J_b = 0: \quad \mathrm{A}:\ \bar{a}_{\mathrm{true}} = 1-e^{-1},\ \ \bar{w}-\bar{w}_s = 1.955 \qquad \mathrm{B}:\ \bar{a}_{\mathrm{true}} = \tfrac{1}{2},\ \ \bar{w}-\bar{w}_s = 1.128 \qquad \mathrm{C}:\ \text{none}$$



truth $= c_{\perp}$ $\quad$ $b_0 = \frac{9}{8}$ (both) $\quad$ $p = 1, \quad \bar{w}_s = 1$

$\hat{b}$ estimated $\quad$ $\sqrt{P[0]} = 0.0157$ (B), $\;$ 0.0708 (C) $\quad$ one seed



truth $= c_{\|}$ $b_0 = \frac{9}{8}$ (both, as on p.6) $p = 1, \quad \bar{w}_s = 1$

$\hat{b}$ estimated $\sqrt{P[0]} = 0.7505$ (B), 0.5697 (C) one seed

$c_{\|}$ substituted into the wall-normal channel; anchor kept, $\sqrt{P[0]} = \sup\left|b_{\text{eff}} - \frac{9}{8}\right|_{c_{\|}}$